

Real Analysis I — Homework 3

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Unless otherwise specified, all functions are defined on $E \subset \mathbb{R}^n$ where $E \in \mathcal{M}$.

For notation simplicity, a set such as $\{x \in E : f(x) > t\}$ is written as $\{f > t\}$ for short.

1. Prove that $f : E \rightarrow \mathbb{R}$ is measurable iff for any $a, b \in \mathbb{R}$ and $a < b$, the set $\{x \in E : a \leq f(x) < b\} \in \mathcal{M}$.
2. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is differentiable. Show that f' is measurable.
3. Suppose $f, g : E \rightarrow \mathbb{R}$ are measurable, and $g \neq 0$ a.e. E . Show that f/g is measurable.
4. Suppose $\{f_k\}$ is a sequence of measurable functions. Show that both $\{x \in E : \lim_k f_k(x) \text{ exists}\}$ and $\{x \in E : \lim_k f_k(x) \text{ does not exist}\}$ are measurable sets. (We consider the limit exists even if $\lim_k f_k = \infty$ or $-\infty$.)
5. Suppose $f : E \rightarrow \bar{\mathbb{R}}$ is measurable and finite a.e. E , where $\mu(E) < \infty$. Show that for any $\delta > 0$, there exists a bounded measurable function $g : E \rightarrow \mathbb{R}$ such that $\mu(\{x \in E : f(x) \neq g(x)\}) < \delta$.
6. Suppose $f_k \xrightarrow{\mu} f$ on E and g is finite a.e. E , where $\mu(E) < \infty$. Show that $f_k g \xrightarrow{\mu} fg$ on E .
7. Suppose $f_k \xrightarrow{\mu} f$ and $g_k \xrightarrow{\mu} g$ on E where $\mu(E) < \infty$, and f, g are finite a.e. E . Show that
 - (a) $f_k + g_k \xrightarrow{\mu} f + g$.
 - (b) $f_k - g_k \xrightarrow{\mu} f - g$.
 - (c) $f_k^2 \xrightarrow{\mu} f^2$.
 - (d) $f_k g_k \xrightarrow{\mu} fg$.
8. Suppose $f : E \rightarrow \mathbb{R}$ and $\phi : \mathbb{R} \rightarrow \mathbb{R}$. Show that if f is measurable and ϕ is monotone (not necessarily strictly), then $\phi(f(x)) : E \rightarrow \mathbb{R}$ is measurable.
9. Suppose $f : E \rightarrow \bar{\mathbb{R}}$ is finite a.e. E . Show that if for any $\delta > 0$, there exists a continuous function $g : E \rightarrow \mathbb{R}$ such that $\mu(\{x \in E : f(x) \neq g(x)\}) < \delta$, then f is measurable.
10. Prove that the converse of Egorov theorem is true for any $E \in \mathcal{M}$ (i.e., $\mu(E) \in [0, \infty]$): if for any $\delta > 0$, there exists a measurable subset $E_\delta \subset E$ such that $\mu(E_\delta) < \delta$ and $f_k \rightrightarrows f$ on $E \setminus E_\delta$, then $f_k \rightarrow f$ a.e. E .